

A Longest Fault-free Cycle in Conditional Faulty Folded Hypercubes*

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Abstract

An n -dimensional folded hypercube, denoted by FQ_n , is an enhanced n -dimensional hypercube with one extra link between nodes that have the furthest Hamming distance. Let FF_v (respectively, FF_e) denote the set of faulty nodes (respectively, faulty links) in FQ_n . Under the assumption that every fault-free node in FQ_n is incident to at least two fault-free links, Hsieh et al. showed that if $|FF_v| + |FF_e| \leq 2n - 4$ for $n \geq 3$, then $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$. In this paper, we show that, under the same conditional fault model, FQ_n with $n \geq 5$ can tolerate more faulty elements and provides the same lower bound of the length of a longest fault-free cycle, i.e., $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$ if $|FF_v| + |FF_e| \leq 2n - 3$.

Keyword: Interconnection networks; Hypercubes; Folded hypercubes; Fault-free cycles; Conditional fault model;

1 Introduction

Hypercubes are powerful networks that can perform various kinds of parallel computations and simulate many other types of networks [18, 21]. Hence, the structural properties of hypercubes have been widely studied [7, 8, 10, 27]. A number of other topologies, such as paths, trees, rings, and meshes, can be embedded into a hypercube. There are also many related results in hypercubes with

faulty elements including node failure or link failure [2, 3, 6, 17, 22]. One of the most popular variants of hypercubes is the so-called *folded hypercube*, which is an extension of the hypercube and can be constructed by adding a link to every pair of nodes with complementary address. The folded hypercube has been shown to be able to improve the system's performance over a regular hypercube in many measurements [1, 13].

Since faults may happen on both nodes and links in a network, it is practically meaningful and important to consider faulty networks. A node is *fault-free* if it is not faulty. A link is *fault-free* if the communication link between its two end-nodes is not faulty. A path (cycle) is *fault-free* if it contains neither faulty nodes nor faulty links. In particular, a *fault-free even cycle* is a fault-free cycle of even length, and a *fault-free Hamiltonian cycle* is a fault-free cycle containing all fault-free nodes in the network. There were two commonly used fault models: *random faults* [9, 23] and *conditional faults* [12, 20]. The former assumed that faults might occur anywhere without any restriction, whereas the latter assumed that faults are subject to some reasonable constraint, such as every fault-free node must be incident with at least two fault-free links for communication. Hereafter, we always assume that every fault-free node is incident to at least two fault-free links when we mention that a network fulfills the conditional fault model.

The problem of fault-tolerant embedding on an n -dimensional folded hypercube FQ_n has been widely studied in [11, 14, 16, 24–26] for random fault model and [15, 19] for conditional fault model. Let FF_v (respectively, FF_e) be the set of faulty nodes (respectively, faulty links) in an n -dimensional folded hypercube FQ_n . In [15], Hsieh et al. have shown the result as Theorem 1.

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Theorem 1 (See [15]). *Under the conditional fault model, if $|FF_v| + |FF_e| \leq 2n - 4$ for $n \geq 3$, then $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$.*

In this paper, we further consider the situation with more faulty elements in FQ_n and prove the following result.

Theorem 2. *Under the conditional fault model, if $|FF_v| + |FF_e| \leq 2n - 3$ for $n \geq 5$, then $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$.*

The rest of this paper is organized as follows. In Section 2, we introduce some necessary definitions and important properties of folded hypercubes. In Section 3, we give the proof of our main result. Finally, a concluding remark is given in Section 4.

2 Preliminaries

Throughout this paper, the terms – network and graph, node and vertex, link and edge – are used interchangeable. An n -dimensional *hypercube* Q_n is also called an n -cube, which can be modeled as a graph with vertex set $V(Q_n)$ and edge set $E(Q_n)$ such that every node is distinctly labeled by an n -bit binary string and two nodes are adjacent provided they differ in exactly one bit. Obviously, there are 2^n nodes and $n2^{n-1}$ links in Q_n . For $u \in V(Q_n)$ with binary string $b_n b_{n-1} \cdots b_2 b_1$, we use $u^{(i)}$, $1 \leq i \leq n$, to denote the binary string $b_n b_{n-1} \cdots b_{i+1} \bar{b}_i b_{i-1} \cdots b_1$, where $\bar{b}_i$ is the one's complement of b_i . Thus, if nodes u and v are adjacent, then $u = v^{(i)}$ and $v = u^{(i)}$ for some $i \in \{1, 2, \dots, n\}$, and we call such a link (u, v) an i -dimensional link. For notational convenience, we also refer to the link $(u, u^{(i)})$ as $\ell_i(u)$ and let $E_i = \{\ell_i(u) | u \in V(Q_n)\}$, i.e., the set consisting of all i -dimensional links in Q_n . Further, let $F(E_i)$ denote the set containing all faulty links of E_i . Clearly, $|E_i| = 2^{n-1}$ for every $1 \leq i \leq n$.

An n -dimensional *folded hypercube* FQ_n can be constructed from an n -cube by adding a link to every pair of nodes with complementary addresses, e.g., node $x = b_n b_{n-1} \cdots b_2 b_1$ and node $\bar{x} = \bar{b}_n \bar{b}_{n-1} \cdots \bar{b}_2 \bar{b}_1$. Thus, FQ_n has 2^{n-1} more links than a regular hypercube Q_n . We call these extra links *skip links* to distinguish them from regular links, and let E_s denote the set containing all skip links in FQ_n . Figure 1 illustrates a 2-dimensional and a 3-dimensional folded hypercubes, respectively.

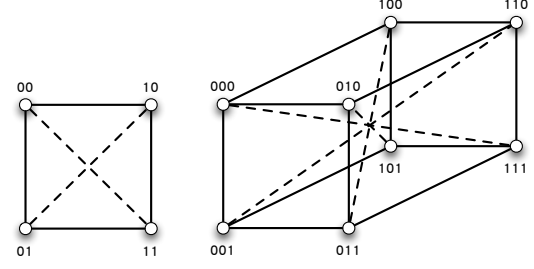


Figure 1: Folded hypercubes FQ_2 and FQ_3 , where skip links are drawn by dashed lines.

In a faulty Q_n or a faulty FQ_n , we let $f(u)$ denote the number of fault-free links incident to a fault-free node u . Also, a node u is called k -free if $f(u) \leq k$. Thus, a node u is 2-free in the conditional fault model if and only if $f(u) = 2$. Clearly, a node u is k -free, then so is $(k+1)$ -free.

Lemma 1. *For $n \geq 3$, if $|FF_v| + |FF_e| \leq 2n - 3$, then there are at most two 2-free nodes contained in FQ_n . In particular, if there are exactly two 2-free nodes contained in FQ_n , then these two nodes must be joined by a faulty link.*

Proof. Clearly, a 2-free nodes in FQ_n is adjacent to at least $(n+1) - 2 = n - 1$ faulty elements (including faulty links and faulty nodes). Assume that FQ_n contains three 2-free nodes u, v and w . Since FQ_n contains no cycle of length three, let us consider a special case that $uv, vw \in E(FQ_n)$ and $uw \notin E(FQ_n)$ to obtain the least total number of faulty elements. Clearly, if $n \geq 3$, the number of faulty elements is $(n-1) * 3 - 2 = 3n - 5 > 2n - 3$, which contradicts to the assumption that $|FF_v| + |FF_e| \leq 2n - 3$. Thus, FQ_n contains at most two 2-free nodes. Furthermore, if there are exactly two 2-free nodes $u, v \in V(FQ_n)$, it is easy to see that u and v must be joined by a faulty link, i.e., the number of faulty elements is $(n-1) + (n-1) - 1 = 2n - 3$. $\square$

We now introduce some known results that will be employed later.

Lemma 2 (See [26]). *There is an automorphism δ of FQ_n such that $\delta(E_i) = E_j$ for $i, j \in \{1, 2, \dots, n\} \cup \{s\}$.*

It directly derives the following corollary.

Corollary 3. *$FQ_n - E_i$ is isomorphic to Q_n for $i \in \{1, 2, \dots, n\} \cup \{s\}$.*

In a faulty hypercube Q_n , we let F_v and F_e be the sets of faulty nodes and faulty links of Q_n , respec-

tively. On the problem of finding the lower bound of a longest fault-free cycle in Q_n , Du et al. [4] and Du and Xu [5] have shown the results as Lemma 4 and Lemma 5, respectively.

Lemma 4 (See [4]). *Under the conditional fault model, the faulty hypercube $Q_n - F_v - F_e$ for $n \geq 3$ contains a fault-free cycle of length at least $2^n - 2|F_v|$ if $|F_v| + |F_e| \leq 2n - 4$ and $|F_e| \leq 2n - 5$.*

Lemma 5 (See [5]). *In Q_n , if $n \geq 5$ and $|F_v| = 2n - 3$, then there is a fault-free even cycle of length at least $2^n - 2|F_v|$.*

In addition, Ma et al. [19] provided the following sufficient condition for a fault-free Hamiltonian cycle in a faulty folded hypercube.

Lemma 6 (See [19]). *If FQ_n ($n \geq 3$) has at most $2n - 3$ faulty links in the conditional fault model, then it contains a fault-free Hamiltonian cycle.*

3 Proof of the main result

In this section, we give the proof of our main result as shown in Theorem 2. We suppose that $f(u) \geq 2$ for every fault-free node $u \in V(FQ_n)$, i.e., every fault-free node in FQ_n is incident to at least two fault-free links. By Theorem 1, if $|FF_v| + |FF_e| \leq 2n - 4$, then the lower bound of the length of a longest fault-free cycle is $2^n - 2|FF_v|$. Thus, we only need to consider the situation $|FF_v| + |FF_e| = 2n - 3$. The proof is divided into two parts. We first consider the situation $|FF_v| \leq n - 5$. Next, we consider the situation $|FF_v| \geq n - 4$, which implies $|FF_e| \leq (2n - 3) - |FF_v| \leq (2n - 3) - (n - 4) = n + 1$.

Lemma 7. *Under the conditional fault model, if $|FF_v| + |FF_e| = 2n - 3$ and $|FF_v| \leq n - 5$ for $n \geq 5$, then $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$.*

Proof. Obviously, the condition $n \geq 5$ is necessary and $|FF_e| = (2n - 3) - |FF_v| \geq (2n - 3) - (n - 5) = n + 2$. By Lemma 1, we may consider the number of 2-free nodes in the following three cases:

Case 1. FQ_n contains no 2-free node. In this case, we have $f(u) \geq 3$ for every node $u \in V(FQ_n)$. Since $|FF_e| \geq n + 2$, there exists a dimension $i \in \{1, 2, \dots, n\} \cup \{s\}$ such that E_i contains at least two faulty links (i.e., $|F(E_i)| \geq 2$). By Corollary 3, $FQ_n - E_i$ is isomorphic to Q_n . Let F_v and F_e be the sets of faulty nodes and faulty links in $FQ_n - E_i$, respectively. It is easy to check that $|F_v| = |FF_v|$ and $|F_e| = |FF_e| - |F(E_i)| \leq (|FF_v| + |FF_e|) -$

$|F(E_i)| \leq (2n - 3) - 2 = 2n - 5$. Moreover, we have $|F_v| + |F_e| = |FF_v| + (|FF_e| - |F(E_i)|) \leq 2n - 5 < 2n - 4$. Since $f(u) \geq 3$ for every node $u \in V(FQ_n)$, it implies that $f(u) \geq 2$ for every node $u \in V(FQ_n - E_i)$. Since $FQ_n - E_i$ is isomorphic to Q_n , by Lemma 4, there exists a fault-free cycle of length at least $2^n - 2|F_v|$ ($= 2^n - 2|FF_v|$) in $(FQ_n - E_i) - F_v - F_e$. Since $(FQ_n - E_i) - F_v - F_e$ is a subgraph of $FQ_n - FF_v - FF_e$, it directly follows that $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$.

Case 2. FQ_n contains a unique 2-free node u . In this case, $f(u) = 2$ and $f(w) \geq 3$ for every other node $w \in V(FQ_n) - \{u\}$. Since $n \geq 5$ and $f(u) = 2$, we may choose a dimension $i \in \{1, 2, \dots, n\} \cup \{s\}$ such that $\ell_i(u)$ is a faulty link. Thus, $|F(E_i)| \geq 1$. By Corollary 3, $FQ_n - E_i$ is isomorphic to Q_n . Let F_v and F_e be the sets of faulty nodes and faulty links in $FQ_n - E_i$, respectively. Then, $|F_v| = |FF_v|$ and $|F_e| = |FF_e| - |F(E_i)| \leq (|FF_v| + |FF_e|) - |F(E_i)| \leq (2n - 3) - 1 = 2n - 4$. Moreover, we have $|F_v| + |F_e| = |FF_v| + (|FF_e| - |F(E_i)|) = (|FF_v| + |FF_e|) - |F(E_i)| \leq (2n - 3) - 1 = 2n - 4$. Since $\ell_i(u)$ is a faulty link, we have $f(u) = 2$ when we consider $u \in V(FQ_n - E_i)$. Also, since $f(w) \geq 3$ for every node $w \in V(FQ_n) - \{u\}$, it implies that $f(w) \geq 2$ for every node $w \in V(FQ_n - E_i) - \{u\}$. We now consider the following two subcases:

Case 2.1: $|F_e| \leq 2n - 5$. Since $FQ_n - E_i$ is isomorphic to Q_n , by Lemma 4, there exists a fault-free cycle of length at least $2^n - 2|F_v|$ ($= 2^n - 2|FF_v|$) in $(FQ_n - E_i) - F_v - F_e$. Since $(FQ_n - E_i) - F_v - F_e$ is a subgraph of $FQ_n - FF_v - FF_e$, it directly follows that $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$.

Case 2.2: $|F_e| = 2n - 4$. Since $|FF_v| + |FF_e| = 2n - 3$ and $|FF_e| = |F_e| + |F(E_i)| \geq (2n - 4) + 1 = 2n - 3$, it implies that $|FF_e| = 2n - 3$ and $|FF_v| = 0$. By Lemma 6, there exists a fault-free Hamiltonian cycle of length $2^n - 2|FF_v| = 2^n$ in $FQ_n - FF_e$.

Case 3. FQ_n contains exactly two 2-free nodes u and v . In this case, $f(u) = f(v) = 2$ and $f(w) \geq 3$ for every other node $w \in V(FQ_n) - \{u, v\}$. By Lemma 1, u and v are connected by a faulty link. Assume without loss of generality that $(u, v) = \ell_1(u) = \ell_1(v)$. Since $n \geq 5$ and $f(u) = f(v) = 2$, there exists a dimension $i \in \{2, 3, \dots, n\} \cup \{s\}$ such that both $\ell_i(u)$ and $\ell_i(v)$ are faulty links. Thus, $|F(E_i)| \geq 2$. By Corollary 3, $FQ_n - E_i$ is isomorphic to Q_n . Let F_v and F_e be the sets of faulty nodes and faulty links in $FQ_n - E_i$, respectively. Then, $|F_v| = |FF_v|$ and $|F_e| = |FF_e| - |F(E_i)| \leq (|FF_v| + |FF_e|) - |F(E_i)| \leq (2n - 3) - 2 = 2n - 5$. Moreover, we have $|F_v| + |F_e| = |FF_v| + (|FF_e| -$

$|F(E_i)|) = (|FF_v| + |FF_e|) - |F(E_i)| \leq (2n-3) - 2 = 2n-5 < 2n-4$. Since $\ell_i(u)$ and $\ell_i(v)$ are faulty links, we have $f(u) = f(v) = 2$ when we consider $u, v \in V(FQ_n - E_i)$. Also, since $f(w) \geq 3$ for every node $w \in V(FQ_n) - \{u, v\}$, it implies that $f(w) \geq 2$ for every node $w \in V(FQ_n - E_i) - \{u, v\}$. Since $FQ_n - E_i$ is isomorphic to Q_n , by Lemma 4, there exists a fault-free cycle of length at least $2^n - 2|F_v|$ ($= 2^n - 2|FF_v|$) in $(FQ_n - E_i) - F_v - F_e$. Since $(FQ_n - E_i) - F_v - F_e$ is a subgraph of $FQ_n - FF_v - FF_e$, it directly follows that $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$. $\square$

Lemma 8. *Under the conditional fault model, if $|FF_v| + |FF_e| = 2n - 3$ and $|FF_e| \leq n + 1$ for $n \geq 5$, then $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$.*

Proof. According to the number of faulty links in FQ_n , we consider the following two cases:

Case 1. $1 \leq |FF_e| \leq n+1$. By Lemma 1, if $n \geq 5$ and there exist two 2-free nodes with a common faulty link, then the number of faulty links in FQ_n is at least $2(n-1) - 1 = 2n - 3 > n + 1$, which contradicts to the assumption that $|FF_e| \leq n + 1$. Thus, we consider the following two subcases when $n \geq 5$:

Case 1.1: FQ_n contains no 2-free node. In this case, we have $f(u) \geq 3$ for every node $u \in V(FQ_n)$. Since $|FF_e| \geq 1$, we may assume that $|F(E_i)| \geq 1$ for some $i \in \{1, 2, \dots, n\} \cup \{s\}$. By Corollary 3, $FQ_n - E_i$ is isomorphic to Q_n . Let F_v and F_e be the sets of faulty nodes and faulty links in $FQ_n - E_i$, respectively. Then, $|F_v| = |FF_v|$ and $|F_e| = |FF_e| - |F(E_i)| \leq (n+1) - 1 = n \leq 2n-5$. Moreover, we have $|F_v| + |F_e| = |FF_v| + (|FF_e| - |F(E_i)|) = (|FF_v| + |FF_e|) - |F(E_i)| \leq (2n-3) - 1 = 2n-4$. Since $f(u) \geq 3$ for every node $u \in V(FQ_n)$, it implies that $f(u) \geq 2$ for every node $u \in V(FQ_n - E_i)$. Since $FQ_n - E_i$ is isomorphic to Q_n , by Lemma 4, there exists a fault-free cycle of length at least $2^n - 2|F_v|$ ($= 2^n - 2|FF_v|$) in $(FQ_n - E_i) - F_v - F_e$. Since $(FQ_n - E_i) - F_v - F_e$ is a subgraph of $FQ_n - FF_v - FF_e$, it directly follows that $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$.

Case 1.2: FQ_n contains a unique 2-free node u . That is, $f(u) = 2$ and $f(w) \geq 3$ for every other node $w \in V(FQ_n) - \{u\}$. Since $n \geq 5$ and $f(u) = 2$, we may choose a dimension $i \in \{1, 2, \dots, n\} \cup \{s\}$ such that $\ell_i(u)$ is a faulty link. Thus, $|F(E_i)| \geq 1$. By Corollary 3, $FQ_n - E_i$ is isomorphic to Q_n . Let F_v and F_e be the sets of faulty nodes and faulty

links in $FQ_n - E_i$, respectively. Then, $|F_v| = |FF_v|$ and $|F_e| = |FF_e| - |F(E_i)| \leq (n+1) - 1 = n \leq 2n-5$. Moreover, we have $|F_v| + |F_e| = |FF_v| + (|FF_e| - |F(E_i)|) = (|FF_v| + |FF_e|) - |F(E_i)| \leq (2n-3) - 1 = 2n-4$. Since $\ell_i(u)$ is a faulty link, we have $f(u) = 2$ when we consider $u \in V(FQ_n - E_i)$. Also, since $f(w) \geq 3$ for every node $w \in V(FQ_n) - \{u\}$, it implies that $f(w) \geq 2$ for every node $w \in V(FQ_n - E_i) - \{u\}$. Since $FQ_n - E_i$ is isomorphic to Q_n , by Lemma 4, there exists a fault-free cycle of length at least $2^n - 2|F_v|$ ($= 2^n - 2|FF_v|$) in $(FQ_n - E_i) - F_v - F_e$. Since $(FQ_n - E_i) - F_v - F_e$ is a subgraph of $FQ_n - FF_v - FF_e$, it directly follows that $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$.

Case 2. $|FF_e| = 0$. Since $|FF_v| + |FF_e| = 2n-3$, we have $|FF_v| = 2n-3$. We arbitrarily choose a dimension $i \in \{1, 2, \dots, n\} \cup \{s\}$ and let F_v be the set of faulty nodes in $FQ_n - E_i$. By Corollary 3, $FQ_n - E_i$ is isomorphic to Q_n that has $|F_v| = |FF_v| = 2n-3$ faulty nodes. By Lemma 5, $(FQ_n - E_i) - F_v$ contains a fault-free even cycle of length at least $2^n - 2|F_v|$ ($= 2^n - 2|FF_v|$). Since $(FQ_n - E_i) - F_v$ is a subgraph of $FQ_n - FF_v$, it directly follows that $FQ_n - FF_v$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$. $\square$

As a consequence, the correctness of Theorem 2 directly follows from Theorem 1 and Lemmas 7 and 8.

4 Conclusion

In this paper, under the conditional fault model (i.e., every fault-free node is incident to at least two fault-free links), we show that if $|FF_v| + |FF_e| \leq 2n-3$ for $n \geq 5$, then $FQ_n - FF_v - FF_e$ contains a fault-free cycle of length at least $2^n - 2|FF_v|$, which is an improvement of the previous result [15]. It is interested to know whether the same result still holds provided there are more faulty elements and a larger n .

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